

AI / ML Quiz - Questions and Answers

Compiled from the provided screenshots. Missing answer details were completed in the same explanation style as the earlier questions.

Q1. What is the key difference between supervised learning and unsupervised learning?

- ☐ A Supervised learning optimizes loss functions, whereas unsupervised learning does not
- ☒ B Supervised learning learns from labels, whereas unsupervised learning does not
- ☐ C Supervised learning can scale to extremely large data size, whereas unsupervised learning cannot
- ☐ D Supervised learning uses deep learning, whereas unsupervised learning does.

Answer: B

Option Explanations

Option A: Incorrect. Many unsupervised methods, such as K-means or autoencoders, also optimize objective functions.

Option B: Correct. Labels are the defining requirement for supervised learning.

Option C: Incorrect. Unsupervised learning often scales well because it does not require human annotation.

Option D: Incorrect. Deep learning techniques can be used in both supervised and unsupervised learning.

Q2. Which of the following statements about the K-means algorithm is incorrect?

- ☐ A K-means algorithm has to know the number of clusters k in advance
- ☐ B K-means algorithm is for unsupervised learning problems
- ☒ C K-means algorithm will find globally optimal clusters in the sense of globally minimizing the sum of point-to-center distances
- ☐ D K-means algorithm provably makes progress at each iteration, until it converges.

Answer: C

Option Explanations

Option A: True statement. 'k' is a pre-defined hyperparameter.

Option B: True statement. It is a fundamental clustering algorithm.

Option C: Incorrect Statement (Correct Choice). K-means is sensitive to initialization and only guarantees a local minimum.

Option D: True statement. The objective function decreases or stays flat at each step.

Q3. Which of the following statements about AI is correct?

- A** AI is about developing machine learning techniques to solve intelligence problems
- B** AI is about developing neural networks and deep learning techniques to solve intelligence problems
- C** AI is about developing computing methods to achieve human-level capabilities and rational behaviors
- D** AI is about developing techniques to make computers behave exactly as humans.

Answer: C

Option Explanations

Option A: Incorrect. ML is only a subset of the broader AI field.

Option B: Incorrect. Deep learning is a subset of ML, which is a subset of AI.

Option C: Correct. AI's core goal is creating systems that act and think rationally to solve complex tasks.

Option D: Incorrect. AI focuses on rational and useful behavior, not necessarily exact human imitation.

Q4. Which of the following is incorrect about supervised learning?

- A** Two major problems of supervised learning are classification and regression
- B** The success of deep learning begins from supervised learning
- C** Many supervised learning algorithms are based on minimizing loss functions
- D** Supervised learning algorithms interact with data samples to identify high-rewarding ones and discard low-rewarding ones, ultimately maximizing overall rewards.

Answer: D

Option Explanations

Option A: True. These are the two primary pillars of supervised learning.

Option B: True. Benchmarks like ImageNet strongly accelerated deep learning through supervised learning.

Option C: True. Loss minimization is the standard learning paradigm in supervised learning.

Option D: Incorrect Statement (Correct Choice). This describes reinforcement learning, not supervised learning.

Q5. Which of the following about relations between linear classifiers and neural networks (NN) is correct?

- A** They are not related
- B** Linear classifiers are a special case of NN, with one hidden layer and indicator function as the activation function

C

Linear classifiers are a special case of NN, with no hidden layer and indicator function as the activation function

D

Linear classifiers are a special case of NN, with no hidden layer and a continuous activation function

Answer: C

Option Explanations

Option A: Incorrect. They share the same fundamental linear algebra basis.

Option B: Incorrect. Adding a hidden layer introduces extra representational structure.

Option C: Correct. A single-layer perceptron with a step or indicator-type output is the textbook form of a linear classifier.

Option D: Incorrect. Using a continuous activation such as sigmoid changes the interpretation and is not the plain textbook linear classifier described here.

Q6. Which of the following statement about neural networks' activation functions is incorrect?

A

They are used to introduce non-linearity to the model

B

Popular choices include ReLU and sigmoid function

C

They need to be differentiable everywhere

D

Choice of activation functions is important to the performance of the network

Answer: C

Option Explanations

Option A: True. Non-linearity allows neural networks to be universal function approximators.

Option B: True. ReLU and sigmoid are standard activation choices.

Option C: Incorrect Statement (Correct Choice). ReLU is not differentiable at $x = 0$, yet it is widely used successfully in practice.

Option D: True. Activation choice affects gradient flow, training speed, and performance.

Q7. Which of the following function can be provably minimized by the gradient descent method?

A

Continuous functions

B

Convex functions

C

Concave functions

D Quadratic functions

Answer: B

Option Explanations

Option A: Incorrect. Continuous but non-convex functions can trap gradient descent in poor local minima or saddle points.

Option B: Correct. In convex optimization, gradient descent has convergence guarantees to a global minimum under suitable conditions.

Option C: Incorrect. Gradient descent moves away from maxima of concave functions rather than solving their natural optimization form.

Option D: Incorrect. Only convex quadratic functions have the relevant guarantee; not every quadratic function is convex.

Q8. Which of the following can be computed during the backpropagation algorithm?

- A** Gradient of the neural network
- B** Predicted value of the neural network
- C** Both the gradient and the predicted value of the neural network
- D** The optimal weights of the neural network

Answer: A

Option Explanations

Option A: Correct. Backpropagation is specifically the efficient computation of gradients using the chain rule.

Option B: Incorrect. Predictions are produced during the forward pass.

Option C: Incorrect. Forward and backward passes are distinct stages; backpropagation itself computes gradients.

Option D: Incorrect. Backpropagation provides gradients, while an optimizer uses them to update weights.

Q9. Which of the following statement about the depth of a neural network (i.e., the number of hidden layers) is correct?

- A** The deeper the network is, the better performance we will obtain
- B** The shallower the network is, the better performance we will obtain
- C** The only issue for training a very deep neural network is the large computational cost
- D** The best choice for the depth of a neural network depends on the amount of available training data and gradient calculation algorithm

Answer: D

Option Explanations

Option A: Incorrect. Excessive depth can lead to optimization difficulties, vanishing gradients, and overfitting.

Option B: Incorrect. Shallow networks may lack the capacity to learn complex hierarchical features.

Option C: Incorrect. Optimization stability and gradient flow are often more serious issues than computation alone.

Option D: Correct. Model depth should be chosen based on data volume, task complexity, and optimization strategy.

Q10. Which of the following is not a major research challenges in deep learning?

- ☐ A Optimizing loss function
- ☐ B Designing neural networks' architecture
- ☒ C Make predictions from learned neural networks
- ☐ D All of the above are very challenging

Answer: C

Option Explanations

Option A: Challenge. Researching better optimizers is a major subfield.

Option B: Challenge. Architecture design is a central driver of deep learning progress.

Option C: Correct Choice. Inference is mainly a deterministic forward pass; engineering it efficiently can be hard, but it is not the same type of fundamental research challenge as optimization or architecture design.

Option D: Incorrect, because option C is not in the same category of major research challenge.

Q11. Which of the following is a drawback of linear classifiers?

- ☐ A Lack of robustness
- ☒ B Have limited expressiveness
- ☐ C Difficult to learn the optimal classifier
- ☐ D None of the above is an issue for linear classifiers

Answer: B

Option Explanations

Option A: Incorrect. Linear models are often quite robust because of their simplicity.

Option B: Correct. They cannot learn nonlinear decision boundaries such as XOR.

Option C: Incorrect. Many linear-model objectives are convex and relatively easy to optimize.

Option D: Incorrect, because option B is a real limitation.

Q12. Why is gradient descent commonly used in training neural networks?

- A** It guarantees finding the global minimum in all cases
- B** It simplifies the network architecture
- C** It provides an efficient iterative method to minimize the loss function
- D** It avoids the need for activation functions

Answer: C

Option Explanations

Option A: Incorrect. For non-convex deep learning objectives, global optimality is not guaranteed.

Option B: Incorrect. Gradient descent optimizes parameters; it does not simplify the architecture itself.

Option C: Correct. It is an efficient iterative method for reducing the loss by updating parameters using gradients.

Option D: Incorrect. Activation functions remain essential for introducing nonlinearity.

Q13. Which of the following problems might arise if the learning rate is set too high?

- A** Slow convergence
- B** Overfitting
- C** The loss may oscillate or diverge
- D** Underfitting

Answer: C

Option Explanations

Option A: Incorrect. Slow convergence is more commonly associated with a learning rate that is too low.

Option B: Incorrect. Overfitting is mainly related to model capacity, data size, and regularization.

Option C: Correct. Overshooting the minimum can cause instability, oscillation, or divergence.

Option D: Incorrect. Although training may fail, oscillation or divergence is the more precise description of what a too-large learning rate causes.

Q14. Which of the following problems might arise if the learning rate is set too high?

- A** Slow convergence
- B** Overfitting
- C** The loss may oscillate or diverge
- D** Underfitting

Answer: C

Option Explanations

Option A: Incorrect. Slow convergence usually happens when the learning rate is too low, not too high.

Option B: Incorrect. Overfitting is primarily determined by model complexity, data, and regularization.

Option C: Correct. If the learning rate is too high, updates can overshoot the minimum and make the loss oscillate or even diverge.

Option D: Incorrect. A too-large learning rate can prevent learning, but the standard and most precise failure mode is oscillation or divergence.